

While other problems occurred, these were the only problems I ended up recording, as their issues were quite prominent. However, all throughout the project, to ensure that the program worked in each of the stages, iterative testing was carried out, to ensure it works. This will be evident in the next section when the final unit tests are carried out.

Unit Tests

For the unit tests, evidence will be displayed below the testing table labelled with figure numbers, along side that, some of the proof will be shown through the video evidence. The time in the video where the evidence appears will be inside of testing evidence column for that particular test.

User Interface (UI) Section

Test No.	Test Description	Testing Evidence	Test Results	Comments
1 (Obj 1)	Main Menu works as intended, displaying the Create New Game, Load Game and Exit button, and each of the buttons can be clicked, and their functions will be complete	Figure 1.	PASS	All 3 buttons are displayed correctly, and all 3 buttons result in the correct outcome (Create button leads to class menu, Load button starts the game with loaded file or shows error message, Exit button quits the application.)
2 (Obj 2)	Create Class menu works as intended, displaying Figher and Archer	Figure 2.	PASS	Both buttons are displayed correctly, and when clicked, the Figher button

	button, and the buttons can be clicked leading to the correct outcome			starts a new game where the player attacks through melee attacks, and the Archer button starts a new game where the player attacks with the ranged attacks
3.1 (Obj 3.1.4)	Player HUD displays the current Level and Room	Figure 3.	PASS	The current game level and current room is displayed on the top right of the screen, and information updates as level and rooms are completed
3.2 (Obj 3.1.1)	Player HUD displays the current Player Item count (Gold and Potion)	Figure 4.	PASS	Player's gold and potion count is displayed on the top right of the screen and gold and potion count updates as player gains more potions and gold.
3.3 (Obj 3.1.2, 3.1.2.1)	Player HUD displays Ability Icons and Ability Cooldowns	Figures 5. and 6.	PASS	Player's current abilities are displayed on the bottom right, displaying an ability icon if the player has the ability, displaying only the number of

				ability icons as the player has (if player has 2 abilities, those 2 icons are displayed, but not the third), and displays an updating cooldown timer instead of the ability icon when the ability is in cooldown.
3.4 (Obj 3.1.3, 3.1.3.1)	Player HUD displays the Player's current Health	Figure 7.	PASS	The player's current health is displayed as a green rectangle on the bottom left of the screen, with a black outline that does not change size, representing the player's maximum health. The green bar decreases in length as the player's loses health matching the player's current health.
3.5 (Obj 3.2)	Player HUD displays Enemy and Object health (Entities)	Figure 8.	PASS	Each entities health is displayed on top of their character inside of the map as a green

				rectangle. The rectangle decreases in size as the entity loses health, until the entity dies and the entity is no longer displayed.
4 (Obj 4)	Pause menu is displayed when user presses ESC, with the Continue and Save and Exit buttons being both displayed and working as intended. (Testing for whether the save actually saves will be on the G.F. Section of testing).	Figure 9.	PASS	Pause menu is accessed by pressing the ESC hotkey, displaying the Continue and Save and Exit Button. Continue button leads back into the game when clicked, Save and Exit returns the user to the Main Menu as intended. (Further testing on the functionality of the Save button will be done when the testing for the Saving is carried out below)
5.1 (Obj 5.1,5.2, 5.3)	Upgrade Screen is displayed when the player kills all of the enemies inside a given room. Displays 3 abilities.	Figure 10.	PASS	In each room, when all of the Enemies are killed, the upgraded screen is displayed with 3 abilities

5.2 (Obj 5.2.2)	If the player already has 3 abilities only the player's abilities are displayed in the shop.	Figures 11 and 12.	PASS	Whenever the player already has 3 abilities, when the upgrade screen is displayed, only the player's abilities are displayed.
6.1 (Obj 6.1, 6.3)	When all of the rooms inside of a given level are completed by killing the enemies inside of them, the shop. Shop can be exited by pressing the hotkey ESC) screen is displayed, displaying 3 item options (Potion, upgrade and Ability). (Tests on item purchases covered later)	Figure 13.	PASS	When all enemies in the last room of the level were killed the shop was displayed, showing Potions in the first slot, Upgrade Player in the middle slot, Ability in the third slot. When the hotkey ESC was pressed, the shop is exited, moving to another screen.
6.2 (Obj 6.2.1, 6.2.2, 6.2.3)	Shop displays each of the item's stock and cost, with the stock updating whenever the player purchase the item.	Figures 14 and 15.	Partial PASS	The costs for all of the items in the shop were displayed, however the stock for the upgrade player and ability items were not displayed. This is not due to a bug,

				but instead the feature was not implemented.
6.3 (Obj 6.2.4)	If the stock of a particular item reaches 0, instead of the item's information being displayed, a sold out message is displayed. Button is also no longer clickable	Figure 16.		When all of a particular item's stock was bought, a SOLD-OUT banner was displayed instead of the item information and button being displayed before. The button could not be clicked either.
7 (Obj 7)	Game Over Screen is displayed when the player's health reaches 0, displaying 2 buttons, Exit to main menu and Quit game. Both buttons work as intended when clicked, with Exit to main menu button resulting in the main menu being displayed, and Quit game resulting in the application being closed.	Figure 17 and 18.	PASS	When the player's health reached below 0, the game stops running and the Game over screen is displayed, displaying the 2 buttons. When the Exit to main menu button is clicked the user is moved to the main menu. When Quit game is clicked, the application is closed.

Game Functions (G.F.) Section

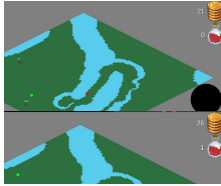
Test No.	Test Description	Testing Evidence	Test Results	Comments
8 (Obj 1)	When a new room starts, the room is displayed in isometric form.	Figure 19.	PASS	Room started and map was displayed in the isometric form.
9 (Obj 1.2, 3.1)	When user clicks a location the player begins to move towards that spot, eventually reaching that spot. If a non-traversable square is clicked nothing happens	Figures 20, 21.	PASS	When a traversable square is clicked the player character moves towards that location. When a non-traversable square is clicked nothing happens.
10 (Obj 1.3)	Whenever a projectile is shot, it moves in a single direction being displayed all the way through.	Figures 22, 23.	PASS	When player and enemy projectiles were shot, the projectiles moved in the direction they were initially projected to,

				being displayed along the way.
11 (Obj 7.2.3, 7.2.4)	When a projectile hits an entity with type different from itself, that entity takes damage.	Figures 24,25,26.	PASS	Every projectile death damage upon collision with a different type of Entity to the one who shot the projectile
12 (Obj 2)	When a new level starts, a collection of rooms is generated to be played through.	Figure 27.	PASS	When a new level started, a new room was displayed, and it was visible that more than 1 room had been added to mapList through the Player HUD.
13 (Obj 1)	When a room is displayed, the room was displayed according to the initial ideas of a particular room, that being 2	Figure 28.	PASS	The room is generated, with green and blue terrain, with blue being non-traversable. It is possible

	different colours/types of terrain, one traversable another non traversable, with the entire room being accessible without the use of a teleport ability.			to see that all parts of the terrain can be accessed by walking there, without any teleport (blink ability).
14 (Obj 2.1)	When a new room start the room has enemies, a player and objects. With Objects spawning anywhere, the player spawning around the centre or corners and enemies spawning away from the player.	Figure 29.	PASS	When the room is loaded in, the enemies spawn far from the player, with the player spawning in one of the corners, and objects spawn in random locations on the map.
15 (Obj 4)	When all enemies in the room are killed the user is moved to the upgrade screen.	Figure 30.	PASS	After all enemies die the upgrade screen is displayed to the player.

16 (Obj 4.1, 4.2)	When the player selects one of the abilities inside of the upgrade screen the ability is added to the player ability list, or the ability selected is upgraded	Figure 31.	PASS	When one ability is selected it is added to the player's ability list, and when the ability selected is already in the player's ability list the ability is upgraded.
17 (Obj 2.2)	When the last room of a given level is completed, by killing all the enemies in that room, the shop screen is displayed.	Figure 32.	PASS	After the level is completed by killing all of the enemies in the last room the shop screen is displayed.
18 (Obj 2.2.(2-7) ,	When the player is in the shop, upon purchasing an item, the stock decreases, the gold is taken away from the player and the player gains the item.	Figures 33, 34, 35.	PASS	After purchasing a health potion, and an upgrade, the player loses the gold of the total cost of the two, the stock decreases for both, and

				the player gains the items.
19 (Obj 3.1)	When the user right clicks a traversable location on the map, the player character moves to that location.	Figures 36, 37.	PASS	After clicking a location on the map, the player character moves to that location.
20 (Obj 3.2)	When the user presses one of the ability hotkeys (1-3), the ability is used, and it is used towards the mouse location if applicable (e.g. bomb/blink), and when the ability finishes it goes on cooldown.	Figures 38, 39.	PASS	When the hotkey is pressed, the ability is used, and when finished it is put on cooldown.
21 (Obj 3.3)	When the user presses the attack hotkey (Q), the player attacks in that direction with the correct class.	Figures 40, 41.	PASS	When the user attacks, with both the fighter and archer class, the attack happens in the direction of the attack,

				either by a projectile appearing and moving in that direction, or the slash attack happening.
22 (Obj 5)	The enemy moves towards the player if any enemy spots the player. Then the enemy will stop a few squares away from the player depending on the class, e.g. Archer stop 9 away, fighter stop 3.	Figures 42, 43.	PASS	The enemies move towards the player when one of the enemies spot the player, and then stops when it is a few squares away from the player.
23 (Obj 6)	When an object is killed, the player gains gold and has chance to get a potion (chance is set at 10%).	Figures 44, 45. 	PASS	(For the sake of the test, the chance to acquire a potion has been set to 100%, so that it can be shown that it does drop a potion) After destroying an

				object the player gains gold and a potion.
24 (Obj 7.2)	When an entity is hit by an attack it takes damage.	Time stamp	PASS	When the player is hit by a slash or projectile it takes damage, and when the enemy and objects are hit by player attacks they also take damage.
25 (Obj 8.3.1)	Blink ability works as intended. (Teleports to the mouse position unless outside of max range, in that case teleports the furthest that way it can go, going on cooldown either case. If the teleport is not valid (to a non-traversable square/outside of the map) a	Time stamp	PASS	Blink ability teleports to the mouse location, unless it is out of range, in that case it teleports to the maximum range that direction, going on cooldown either case. If final teleport location is invalid, the ability simply displays a

	black circle is displayed in the location it tried to teleport and does not allow for the teleport.			black circle and does not go on cooldown.
26 (Obj 8.3.2)	Speed up ability speeds the player up when activated, then goes into cooldown.	Time stamp	PASS	When activated the speed up ability increases the player speed, shown by the fact that the player crossed the same space in less time.
27 (Obj 8.3.3)	Aura ability displays a circle around the player and deals damage periodically when enemies are inside of the circle, going on cooldown after the active time is over.	Time stamp	PASS	When the aura ability is activated, a white circle is displayed around the player, and the enemies inside of the circle take damage periodically.
28 (Obj 8.3.4)	When activated, the stop ability	Time stamp	PASS	When the ability is activated,

	stops all enemy actions, and then after an active time, it goes on cooldown, with the enemies going back to normal.			the enemies stop moving and stop attacking. Then after some time the enemies go back to moving and attacking and the ability goes on cooldown.
29 (Obj 8.3.5)	When the Bomb ability is activated a bomb (black) projectile is shot to the mouse's location. If the projectile collides with an entity, it explodes and deals damage to enemies, if it does not collide before it reaches the mouse location, it explodes where the projectile was initially shot towards. The	Time stamp	PASS	When the Bomb ability is activated, the projectile is shot, and when it collided with an enemy it exploded and dealt damage. When it didnt collide with the enemy it moved towards the location it was initially shot towards and exploded there.

	ability then goes on cooldown			
30 (Obj 8.4.1)	Blink ability scaling works as intended, with the range increasing and the cooldown decreasing.	Time stamp	PASS	After the blink ability is upgraded, the cooldown decreases and the range can be seen to have increased.
31 (Obj 8.4.2)	Speed Up ability scaling works as intended, with the speed increase increasing, active time increasing, and the cooldown decreasing	Time stamp	PASS	When upgraded the active time is longer, with the speed increase also being visibly bigger, with the cooldown also decreasing.
32 (Obj 8.4.3)	Aura ability scaling works as intended, with the radius increasing, the damage being applied more often, and more damage being applied, on top of active time	Time stamp	PASS	When upgraded the ability's status all increase/ decrease as they should, as per the test description.

	increasing and the cooldown decreasing.			
33 (Obj 8.4.3)	Stop ability scaling works as intended, with the active time increasing, and the cooldown decreasing.	Time Stamp	PASS	When the ability is upgraded the active time increases and the cooldown decreases.
34 (Obj 8.4.5)	Bomb ability scaling works as intended, with the bomb projectile size increasing and the damage dealt also increasing, on top of the cooldown decreasing.	Time Stamp	PASS	When the bomb ability is upgraded, the bomb radius and damage increase, and the cooldown decreases.
35 (Obj 9)	When the Save and Exit option is selected, the user is able to save the game and go back to the game's state before the save. This should only save the current level in terms of the	Time stamp	PASS	The user is able to save the game, and return to the same level with the same health, abilities, gold and potion count, regardless of whether the application

	level and room state, however gold potions, health and abilities should all remain the same.			has been closed or not.
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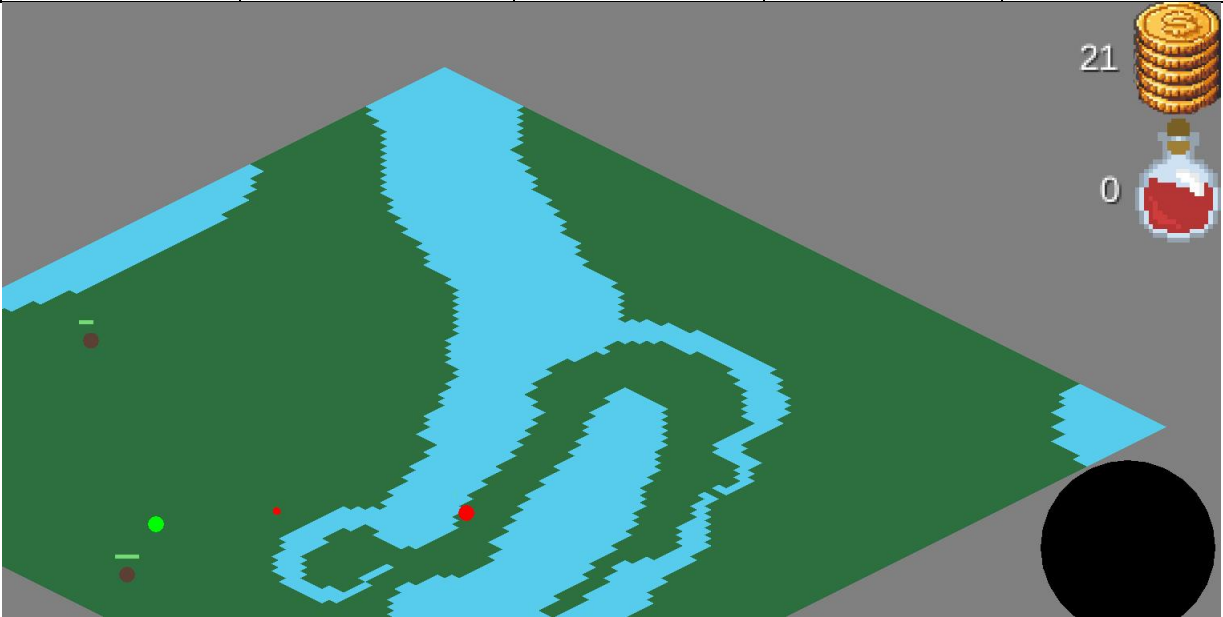


Figure 44 showing player has 21 gold and 0 potions, with the objects on screen (brown circles)

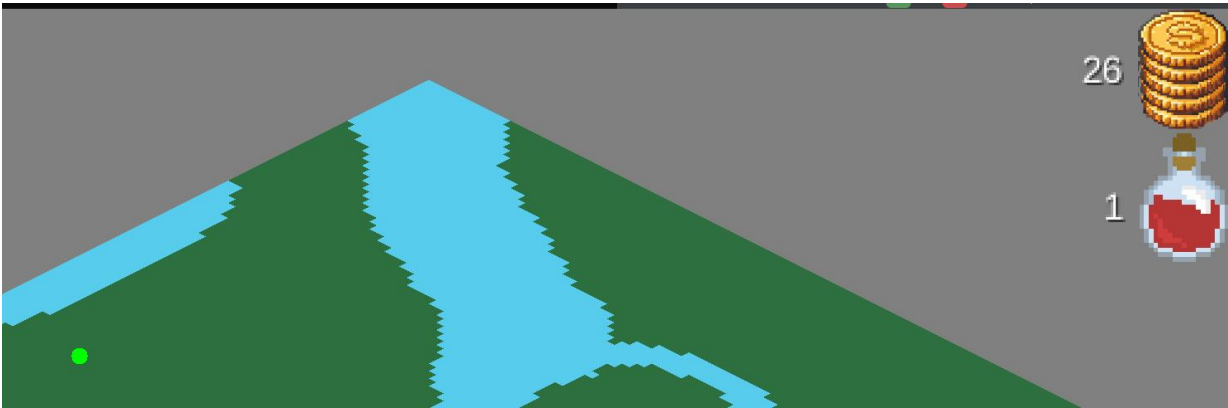


Figure 45 showing player now has 26 gold and 1 potion, with the object gone.